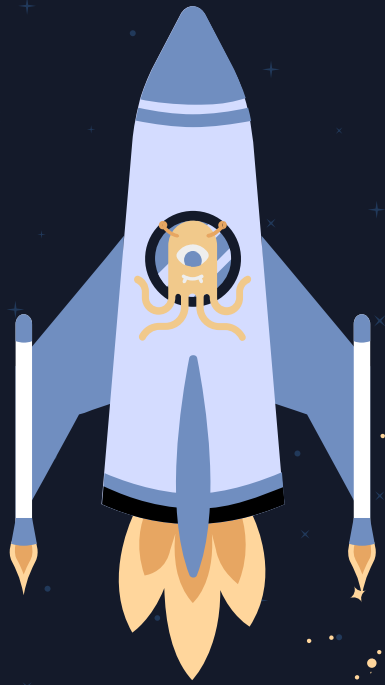
Does the axial ratio of Spiral and Elliptical Galaxies affect the color? If yes, how?

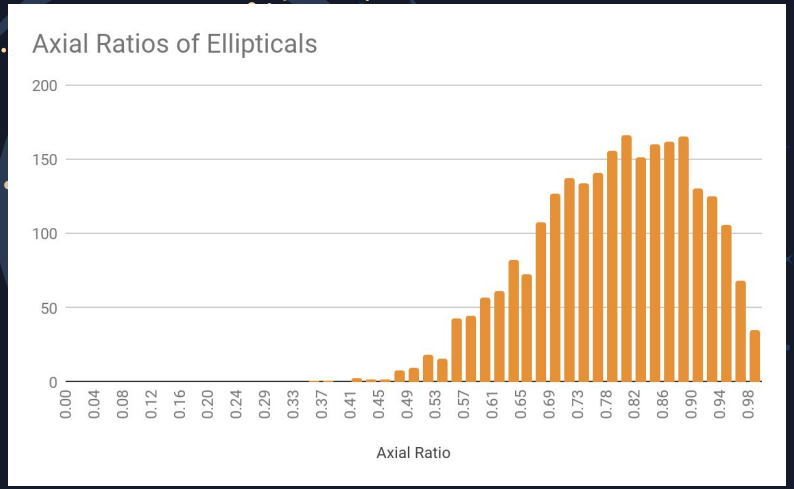
Meghan and Aidan

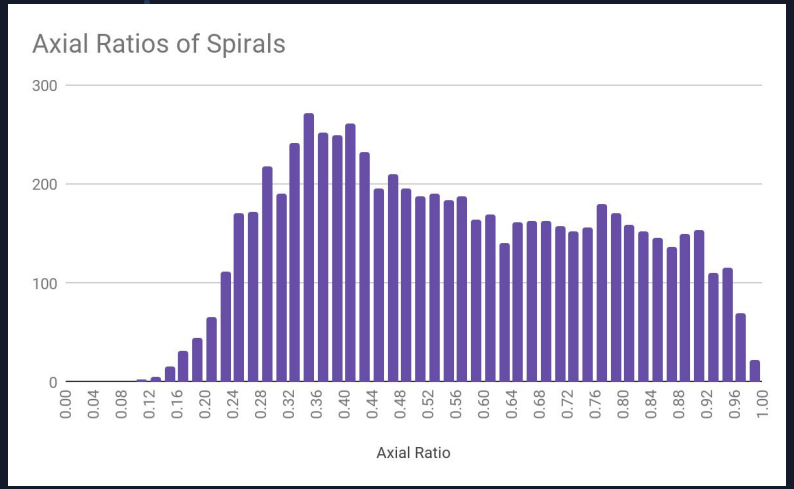
Important Terms

Axial Ratio: tells you how round the galaxy is. A value of 1 tells you it is round and if a spiral galaxy has a value of <= 0.2, the galaxy could be viewed edge on.

Determining Color: GMAG - RMAG (G-R)





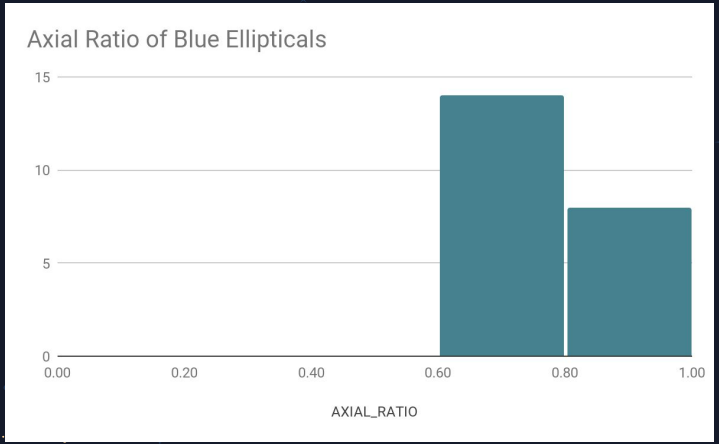


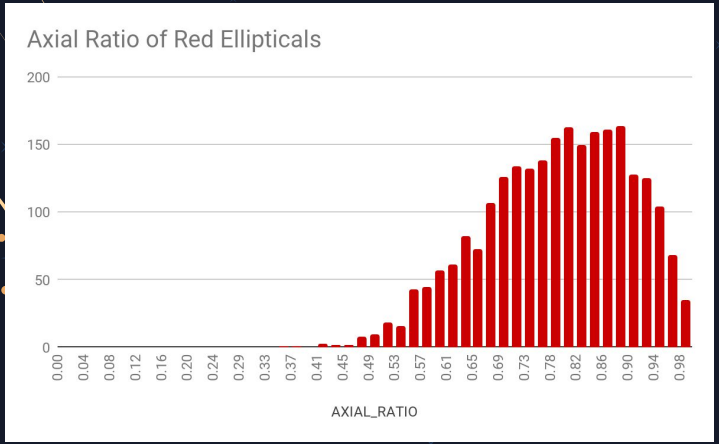
Comparing Axial Ratios

- Split data into ellipticals and spiral galaxies using Galaxy Zoos volunteer data

- Ellipticals tend to be more round having a higher value for axial ratio
  - Median: .80
  - Mean: .79

- Spirals tend to be more scattered in size, varies how the galaxy is angled towards us.
  - Median: 0.53
  - Mean: 0.56

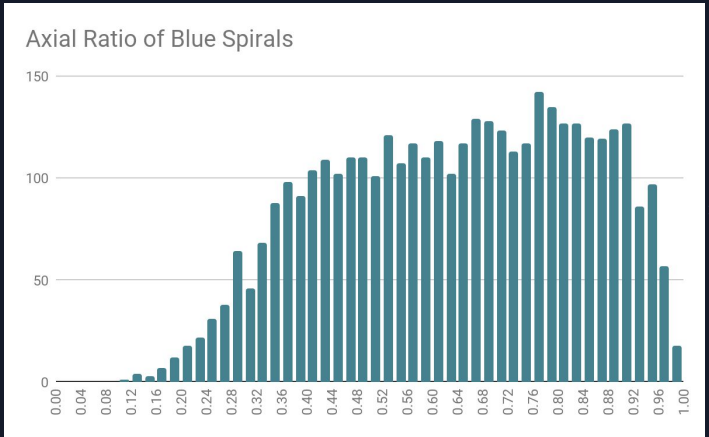


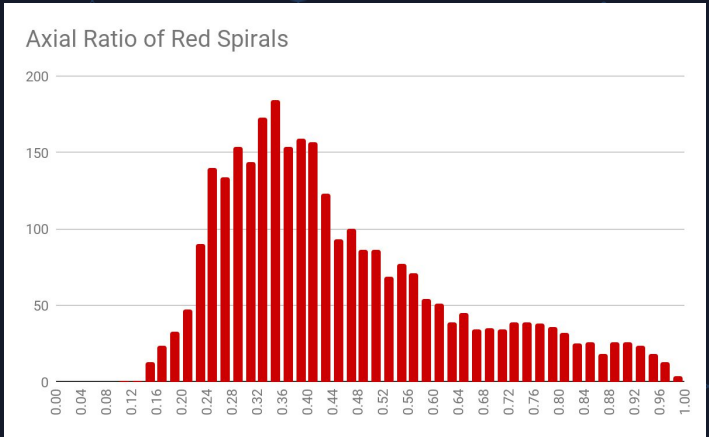


Red and Blue Ellipticals

- We further split the data by color, red and blue.
- Ellipticals appear to have more reddish galaxies and less in blue.
- Avg. Axial ratio of blue ellipticals: .80
- Avg. Axial ratio of red ellipticals: .79

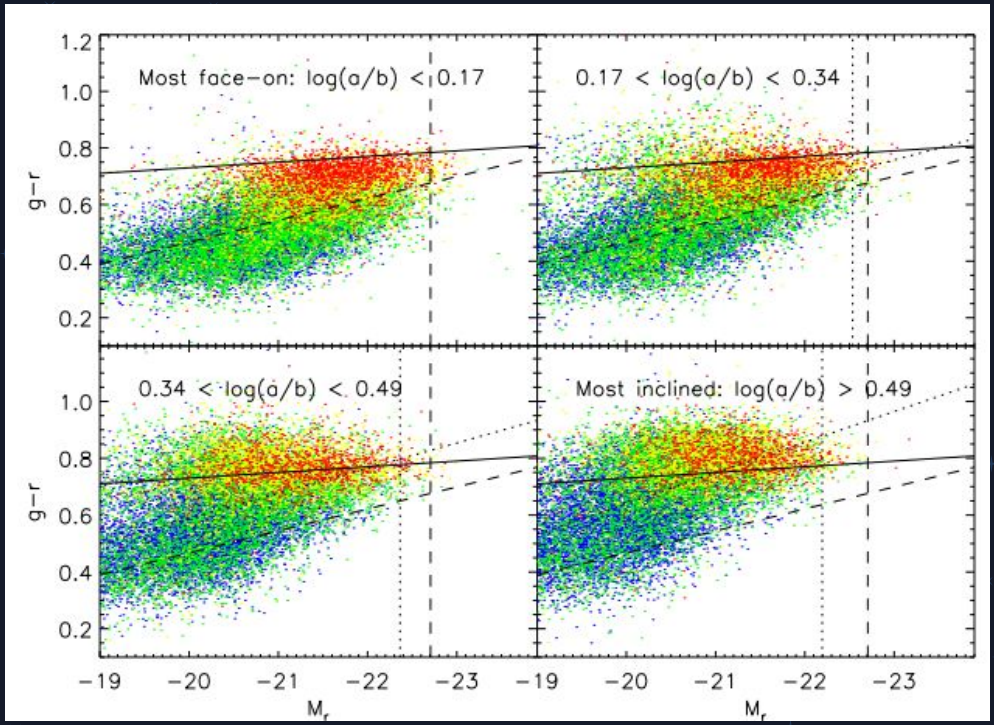
Blue and Red Spirals



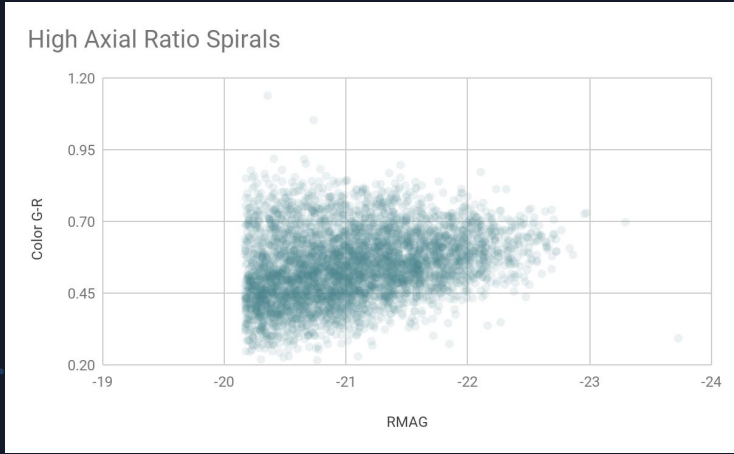


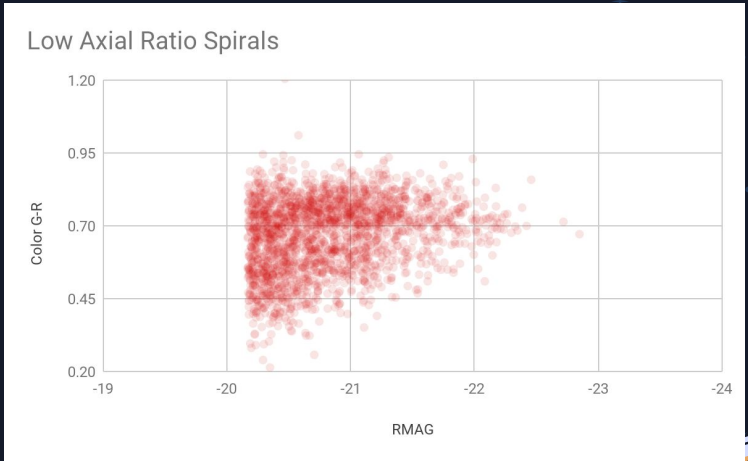
- Spirals galaxies that are less round appear reddish. While rounder spiral galaxies tend to be more blue.
- Avg. Axial ratio of Red Spirals: .45
- Avg. Axial ratio of Blue Spirals: .64

Color Magnitude Plot



Our interpretation of the Color Magnitude Plot





What did we learned?

Does the axial ratio of Spiral and Elliptical Galaxies affect the color?

Yes, for spiral galaxies

View on edge = more dust

View face on = newer, blue stars

Sources:

https://arxiv.org/pdf/1001.1744.pdf

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